

Chapter

5

Work

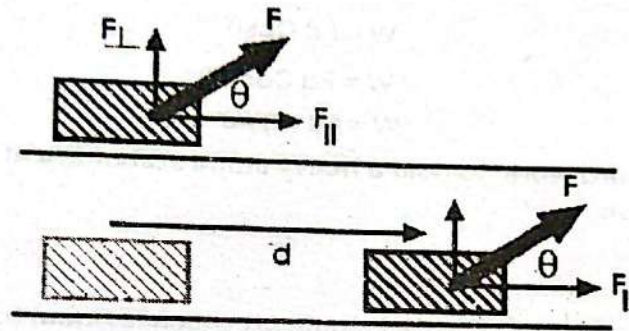
DEFINITION

"Work is said to be done when a force displaces a body in its particular direction"

OR

"The product of force ' $\vec{F}$ ' in the direction of displacement ' $\vec{d}$ ' and the magnitude of the displacement".

WORK DONE BY CONSTANT FORCE:



Work is a scalar quantity by definition and is given by the dot product of force $\vec{F}$ and displacement $\vec{d}$, i.e.

$$W = \vec{F} \cdot \vec{d}$$

Since the force vector $\vec{F}$ and the displacement vector $\vec{d}$ are in same direction, therefore

$$W = Fd \cos \theta$$

Work is a scalar quantity. It can be positive or negative depending on the value of angle between force $\vec{F}$ and displacement.

TYPES OF WORK:

Positive Work:

When the component of force is in the same direction of displacement, i.e. $\theta = 0^\circ$ then work is positive.

$$W = Fd \cos \theta$$

$$W = Fd \cos (0)$$

$$[\because \cos (0) = 1]$$

$$W = Fd (1) = Fd$$

Example: When a spring is stretched the work done by the stretching force is positive.

Negative Work

When the direction of force is opposite to the direction of displacement, i.e. $\theta = 180^\circ$, then work is negative.

$$W = Fd \cos(\theta)$$

$$W = Fd \cos 180^\circ$$

$$W = Fd (-1) = -Fd$$

$$[\because \cos 180 = -1]$$

Example: The work done by the gravitational force on the body being lifted is negative. Since the upward displacement is opposite to the gravitational force.

Zero Work

When the force acts at right angles or $\perp$ to the displacement, i.e. $\theta = 90^\circ$, then work is zero.

$$W = Fd \cos(\theta)$$

$$W = Fd \cos 90^\circ$$

$$W = Fd (0) = 0$$

$$[\because \cos 90 = 0]$$

Example: It is considered 'hard work' to hold a heavy stone stationary at stretched hand but no work is done in the technical sense.

Units Of Work

- In M.K.S system:
In S.I units, unit of work is joule (J) which is equal to N - m.
- In C.G.S System
In C.G.S Systems, unit of work is erg which is equal to Dyne .cm .
- In F.P.S System
In F.P.F Systems , unit of work is ft -lb .

Other Units

There is another unit of work done that is British thermal unit (B.T.U)

$$1055 \text{ J} = 1 \text{ BTU}$$

For energy of subatomic particles it is measured in eV.

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

The multiples of electron volt are

- Million electron-volt = 1 MeV = 10^6 eV
- Tera electron-volt = 1 TeV = 10^{12} eV

WORK DONE BY CONSTANT DISPLACEMENT TIME GRAPH:

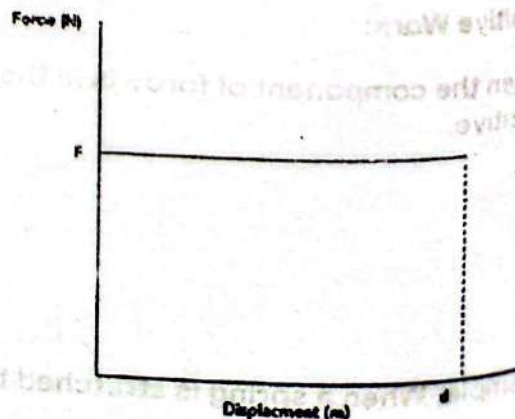
The Area under the curve of a FORCE-DISPLACEMENT GRAPH gives the WORKDONE where Force acting on an object is taken on y-axis and displacement is on x-axis.

CONSTANT FORCE

When the force acting on a system is constant, e.g. the force of gravity (weight of an object), or friction when moving at a constant speed, the force-displacement graph is just a horizontal line because the force does not change.

In this case, the area under the graph is just a rectangle, with a base of length s and a height of length F . The area of the rectangle is base $\times$ height, so,

$$W = F.d$$

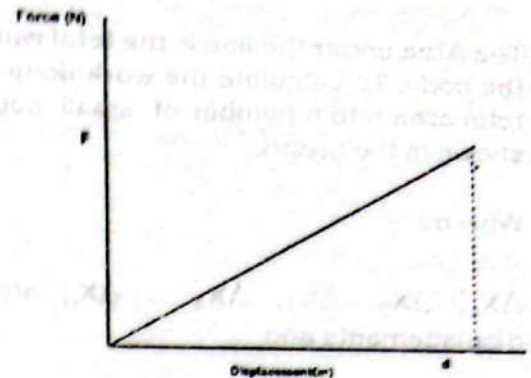


NON-CONSTANT FORCE:

In this case, the area under the graph makes a triangle so we use the formula for the area of a triangle ($\frac{1}{2} \times \text{base} \times \text{height}$) to calculate the work done. The base of the triangle is the extension, e . The height of the triangle is the force when extended by d ,

So the work done is,

$$W = \frac{1}{2} F \cdot d$$



Chapter 5

Work by Variable Force and Kinetic Energy

WORK DONE BY VARIABLE FORCE:

The work done by a variable force is the amount of energy transferred or expended when a force, which can vary in magnitude and direction with respect to position or displacement, acts on an object and causes it to move over a certain distance.

Consider a particle which moves along a curved path subject to a variable force.

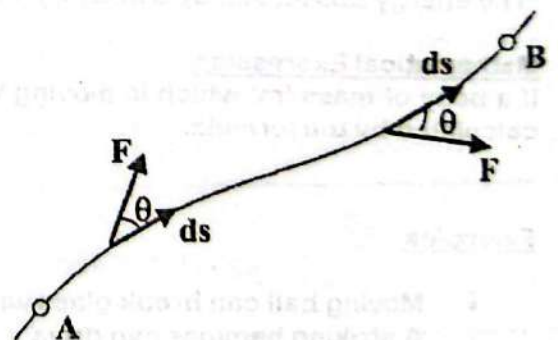
The work done can be written as :

$$dW = \vec{F} \cdot d\vec{s}$$

The Total Work done from point A to B will be:

OR Generally $W_{A \rightarrow B} = \sum F \cdot \Delta s$

$$W_{A \rightarrow B} =: \int_A^B \vec{F} \cdot d\vec{s} = \int_A^B (F \cos \theta) ds$$



WORK DONE BY VARIABLE FORCE (GRAPHICAL REPRESENTATION):

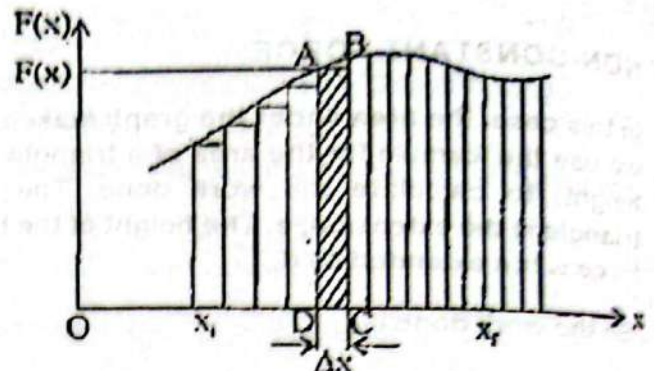
Consider a body covers displacement from x_i to x_f , when a variable force acts on it.

The Area under the line is the total workdone by the body. To calculate the work done we divide total area into n number of small segments as shown in the graph.

Where :

$\Delta x_1, \Delta x_2, \Delta x_3, \Delta x_4 \dots \Delta x_n$ are covered displacements and

$F_{x1}, F_{x2}, F_{x3}, F_{x4} \dots F_{xn}$ are their corresponding Forces.



For Total Work Done we can say that

$$W = W_1 + W_2 + W_3 + W_4 + \dots + W_n$$

For Total Work Done we can say that

$$W = \sum_{i=1}^n W_i$$

$$W = \vec{F}_1 \cdot \vec{\Delta x}_1 + \vec{F}_2 \cdot \vec{\Delta x}_2 + \vec{F}_3 \cdot \vec{\Delta x}_3 + \vec{F}_4 \cdot \vec{\Delta x}_4 + \dots + \vec{F}_n \cdot \vec{\Delta x}_n = \sum_{i=1}^n \vec{F}_{xi} \cdot \vec{\Delta x}_i$$

The above is the equation of total workdone when a variable force acts on it.

KINETIC ENERGY:**Definition**

"The energy possessed by a body by virtue of its motion is called kinetic energy".

Mathematical Expression

If a body of mass 'm' which is moving with velocity of 'n' then the kinetic energy of the body can be calculated by the formula.

$$K.E = \frac{1}{2}mv^2$$

Examples

- Moving ball can break glass window.
- A striking hammer can drive a nail.

DERIVATION OF THE FORMULA OF KINETIC ENERGY

To find the expression for K.E of an object in motion, we have to determine the work done by the moving object. This work is obviously equal to the change in K.E of the object.

Consider a body of mass 'm' which is projected up in the gravitational field with the velocity 'n'. After attains the maximum height 'h' the body comes to rest. The initial kinetic energy of the body is capable of doing work and is used up in doing work against the force of gravity.

At the maximum height the kinetic energy of the body is zero, because it is no more capable of doing work against the gravity. This means that the total work done by the body is a measure of its initial kinetic energy.

Work done by the body = $\vec{F} \cdot \vec{S}$

$$W = FS \cos \theta$$

Since $\theta = 0^\circ$, Therefore

$$W = FS \cos 0^\circ = FS$$

But the distance covered in vertical direction is equal to height ' h ', and $F = \text{Weight} = mg$

$$W = mgh$$

Now we calculate ' h ' by using the following data

Initial velocity of the body = v_i

Final velocity of the body = $v_f = v = 0$

Acceleration of the body = $a = -g$

Distance = $S = h = ?$

$$\begin{aligned} 2aS &= v_f^2 - v_i^2 \\ 2(-g)h &= (0)^2 - v^2 \\ -2gh &= -v^2 \end{aligned}$$

$$W = mg \cdot \frac{v^2}{2g}$$

$$W = \frac{mv^2}{2}$$

$$W = \frac{1}{2}mv^2$$

Hence the Kinetic Energy of the body of mass ' m ' and moving with velocity ' v ' is

$$K.E = \frac{1}{2}mv^2$$

Chapter

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Potential Energy

POTENTIAL ENERGY:

Definition

"The energy possessed by a body by virtue of its position is called potential energy".

Mathematical Expression

If a body of mass ' m ' is lifted to a height ' h ' then the Potential of the body can be calculated by the formula.

$$P.E = mgh$$

GRAVITATIONAL POTENTIAL ENERGY:

Gravitational potential energy is the energy stored in an object when work is done against gravitational force.

When work is done on a body against the gravitational force of an earth from y_1 level to y_2 level:

$$W_{\text{grav}} = Fs = -mg(y_1 - y_2)$$

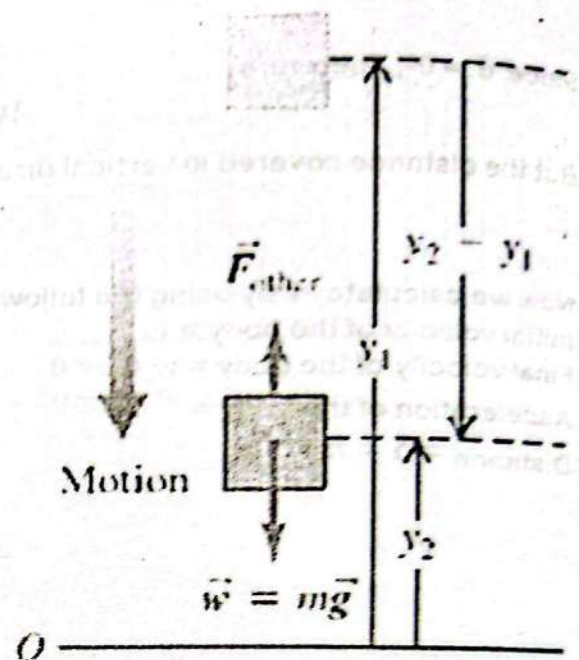
$$W_{\text{grav}} = mgy_2 - mgy_1 = mg(y_2 - y_1)$$

$$W_{\text{grav}} = mgh \quad : y_2 - y_1 = h$$

Hence :

$$\text{P.E} = mgh$$

Thus, potential energy of a body in the earth's gravitational field at a height "h" is "mgh" which is positive quantity with respect to earth's surface which is supposed to be the level of an arbitrary zero potential.

**WORK DONE AGAINST GRAVITATIONAL FIELD:**

we consider a ball of mass 'm' which is taken very slowly to the height 'h'. The very slow motion is possible only when the applied force on the body by an external agency is equal in magnitude to that of the force of gravity, i.e.

The work done by the applied force is

$$W = \vec{F} \cdot \vec{S} = FS \cos \theta$$

Since $S = h$ and $\theta = 0$, Therefore

$$W = Fh \cos(0)$$

$$W = mgh(1)$$

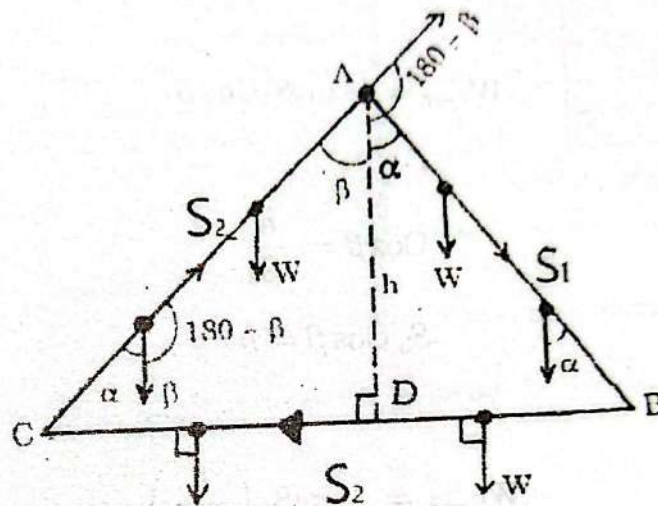
$$W = mgh$$

Thus the work done on a body by applying an external force against the gravitational force is stored in it in the form of potential energy.

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**Work Done in Gravitational Field is
Independent of Path**

To prove this statement, we consider a closed path of any shape in the gravitational field and show that the work done in carrying a body along this path is zero. For the sake of simplicity we take a triangular path ABCA in which the base BC is perpendicular to the gravitational field as shown in figure. The amount of work done in carrying the body from A to B, B to C and from C to A are represented by $W_{A \rightarrow B}$, $W_{B \rightarrow C}$, and $W_{C \rightarrow A}$ respectively.



WORK DONE FROM A $\rightarrow$ B
In $\triangle ABD$

$$W_{A \rightarrow B} = \vec{F} \cdot \vec{S}_1 = (F)(S_1 \cos \alpha) \quad \text{-----eq.(1)}$$

$$\frac{AD}{AB} = \cos \alpha$$

$$\frac{h}{S_1} = \cos \alpha$$

$$h = S_1 \cos \alpha$$

By substituting the value of h in equation (1)

$$W_{A \rightarrow B} = mgh$$

WORK DONE FROM B $\rightarrow$ C

$$W_{B \rightarrow C} = \vec{F} \cdot \vec{S}_2 = F(S_2 \cos 90) = FS_2(0) = 0$$

$$W_{B \rightarrow C} = 0$$

WORK DONE FROM C $\rightarrow$ A

$$W_{C \rightarrow A} = \vec{F} \cdot \vec{S}_3$$

$$W_{C \rightarrow A} = FS_3 \cos(180^\circ - \beta)$$

$$\text{As } [\because \cos(180^\circ - \beta) = -\cos \beta]$$

$$W_{C \rightarrow A} = -FS_3 \cos \beta$$

Where $F = mg$

$$W_{C \rightarrow A} = -mgS_3 \cos \beta$$

In $\triangle CAD$

$$\cos \beta = \frac{h}{S_3}$$

$$S_3 \cos \beta = h$$

Therefore,

$$W_{C \rightarrow A} = -mgS_3 \left(\frac{h}{S_3} \right)$$

$$W_{C \rightarrow A} = -mgh$$

Total work done along the closed path ABCA

$$W_{A \rightarrow B} + W_{B \rightarrow C} + W_{C \rightarrow A} = mgh + 0 - mgh = 0$$

Thus whether we carry the body from A to C (along AC directly) or along the path ABC, the work done is the same. There may be an infinite number of paths going from A to C, but the work done along any path is the same. Such a type of field of force in which the work is independent of the path is called a conservative field. Thus gravitational field is a conservative field.

Chapter 5

Absolute Gravitational Potential Energy

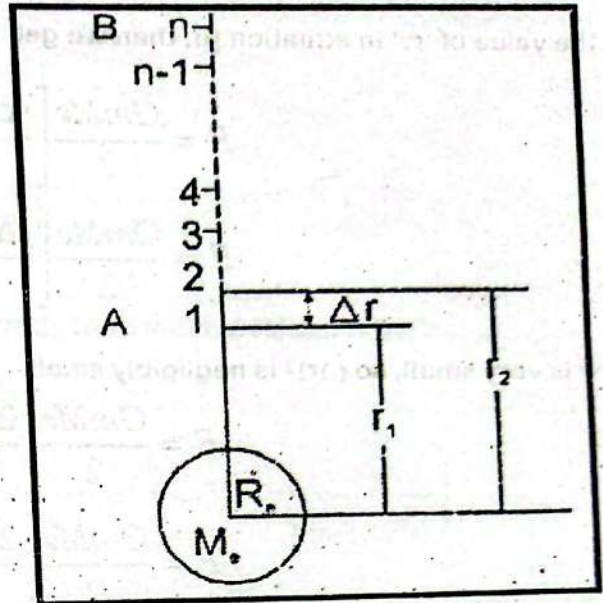
Absolute gravitational potential energy is the amount of work that needs to be done to bring an object from a point of infinite distance to its current location.

DERIVATION:

In order to calculate the absolute potential energy of the body, we divide the entire displacement into a large number of small displacement intervals and applying Newton's law of gravitation.

Suppose a point B is situated at a very large distance from the surface of the earth in the gravitational field as shown in figure.

Now consider a body of mass 'm' from an initial position A (or 1) to the final position B (or n). We divide the distance between A and B into a large number of intervals of equal small width Δr each.



Since Δr is small, the force of gravity throughout this interval may be assumed to be constant.

The magnitude F_1 of the force $\vec{F}_1$ acting at the point 1 is given by

$$F_1 = \frac{GmM_e}{r_1^2}$$

Where M_e is the mass of earth, G is the gravitational constant and r_1 is the distance of point 1 from the center of earth.

Similarly the magnitude F_2 of the force $\vec{F}_2$, acting at point 2 is given by

$$F_2 = \frac{GmM_e}{r_2^2}$$

The average force acting throughout the first interval.

$$F = (F_1 + F_2)/2$$

Where F represents the magnitude of the average force $\vec{F}$, therefore

$$F = \frac{1}{2} (F_1 + F_2)$$

$$F = \frac{1}{2} \left[\frac{GmM_e}{r_1^2} + \frac{GmM_e}{r_2^2} \right]$$

$$F = \frac{GmM_e}{2} \left[\frac{r_2^2 + r_1^2}{r_1^2 r_2^2} \right] \text{-----(i)}$$

Since $r_2 - r_1 = \Delta r$ from figure then $r_2 = \Delta r + r_1$

put the value of ' r_2 ' in equation (i), then we get

$$F = \frac{GmMe}{2} \left[\frac{(\Delta r + r_1)^2 + r_1^2}{r_1^2 r_2^2} \right]$$

$$F = \frac{GmMe}{2} \left[\frac{(\Delta r)^2 + 2\Delta r r_1 + r_1^2 + r_1^2}{r_1^2 r_2^2} \right]$$

As Δr is very small, so $(\Delta r)^2$ is negligibly small.

$$F = \frac{GmMe}{2} \left[\frac{2\Delta r r_1 + 2r_1^2}{r_1^2 r_2^2} \right]$$

$$F = \frac{GmMe}{2} \left[\frac{2r_1(\Delta r + r_1)}{r_1^2 r_2^2} \right]$$

$$F = \frac{GmMe}{2} \left[\frac{2r_1 r_2}{r_1^2 r_2^2} \right]$$

$$F = \frac{GmMe}{r_1 r_2} \quad \text{--- (ii)}$$

The work done in lifting the body from point 1 (position A) to point 2, by an applied force, is equal and opposite to the average gravitational force is given by

$$W_{12} = \vec{F} \cdot \vec{\Delta r}$$

Since the applied force and the displacement in same direction, therefore

$$W_{12} = F \Delta r \cos 0$$

i.e. $\theta = 0$ and $\cos(0) = 1$

$$W_{12} = F \Delta r (1)$$

$$W_{12} = F \Delta r$$

Substitute the value of ' F ' from equation (ii) and Δr

$$W_{12} = \frac{GmMe}{r_1 r_2} \cdot (r_2 - r_1)$$

$$W_{12} = GmMe \left[\frac{r_2 - r_1}{r_1 r_2} \right]$$

$$= GmMe \left[\frac{r_2}{r_1 r_2} - \frac{r_1}{r_1 r_2} \right]$$

$$W_{12} = GmMe \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

The above equation shows the W_{12} is the work done in lifting the body from point 1 to point 2.
Similarly the work done in lifting the body from point 2 to point 3 is

$$W_{23} = GmMe \left[\frac{1}{r_2} - \frac{1}{r_3} \right]$$

And the work done from the point $(n-1)$ to n is

$$W_{(n-1)n} = GmMe \left[\frac{1}{r_{n-1}} - \frac{1}{r_n} \right]$$

Hence the total work done by the applied force in lifting the body from initial position A to final position B, we get,

$$W = W_{12} + W_{23} + \dots + W_{(n-1)n}$$

$$W = GmMe \left(\frac{1}{r_1} - \frac{1}{r_2} \right) + GmMe \left(\frac{1}{r_2} - \frac{1}{r_3} \right) + \dots + GmMe \left(\frac{1}{r_{n-1}} - \frac{1}{r_n} \right)$$

$$W = GmMe \left[\frac{1}{r_1} - \frac{1}{r_2} + \frac{1}{r_2} - \frac{1}{r_3} + \frac{1}{r_3} - \frac{1}{r_4} + \dots + \frac{1}{r_{n-1}} - \frac{1}{r_n} \right]$$

$$W = GmMe \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

This is the potential energy, which is represented by U of the body at the point B with respect to point A.

Hence the potential energy of the body at the point A with respect to that to the point B is $\Delta U = -W$

$$\therefore \Delta U = -GmMe \left(\frac{1}{r_1} - \frac{1}{r_n} \right)$$

$$\text{Or } \Delta U = GmMe \left(-\frac{1}{r_1} + \frac{1}{r_n} \right)$$

$$\text{Or } \Delta U = GmMe \left(\frac{1}{r_n} - \frac{1}{r_1} \right)$$

Where the point B lies at an infinite distance, i.e. $r_n = \infty$ the potential energy at that point is zero, then

$$\Delta U = U$$

$$U = (P.E_{abs}) = GmEm \left(\frac{1}{\infty} - \frac{1}{r_1} \right)$$

$$U = P.E_{abs} = GmEm \left(0 - \frac{1}{r_1} \right)$$

$$U = P.E_{abs} = GmEm \left(-\frac{1}{r_1} \right)$$

$$U = P.E_{abs} = \frac{-GMEm}{r_1} - (v)$$

Assigning an arbitrary value of r , i.e. ($r_1 = r$) in equation (v)

$$U = P.E_{abs} = -\frac{GMEm}{r}$$

Therefore the absolute P.E of a body of mass m lying at the surface of the earth is given by
Where ' R_e ' is the radius of the earth.

$$P.E_{abs} = -\frac{GMEm}{R_e}$$

And the negative sign indicates that the potential energy is negative at any finite distance; i.e. the potential energy is zero at infinity and decreases as the separation distance decreases.
Thus "The fact that the gravitational force acting on the particle by the earth is attractive".

Value of the absolute potential energy at a height 'h':

An approximate value of the absolute potential energy at a height 'h' i.e.

($h \ll R_e$) above the surface at the earth can be obtained from the last equation

$$P.E_{abs} = -\frac{GMEm}{R_e}$$

$$P.E_{abs} = -\frac{GMEm}{R_e + h}$$

$$P.E_{abs} = -\frac{GMEm}{R_e \left(1 + \frac{h}{R_e}\right)}$$

$$P.E_{abs} = -\frac{GMEm}{R_e} \left(1 + \frac{h}{R_e}\right)^{-1}$$

The expression $\left(1 + \frac{h}{R_e}\right)^{-1}$ can be expanded by using the binomial theorem i.e.

$$(a + b)^n = a^n + n a^{n-1} b^1 + \dots$$

$$\left(1 + \frac{h}{R_e}\right)^{-1} = (1)^{-1} + (-1)(1)^{-1-1} \left(\frac{h}{R_e}\right)^1$$

$$\left(1 + \frac{h}{R_e}\right)^{-1} = 1 - \frac{h}{R_e}$$

Where we have neglected the higher order terms and therefore

$$P.E_{abs} = -\frac{GMEm}{R_e} \left(1 - \frac{h}{R_e}\right)$$

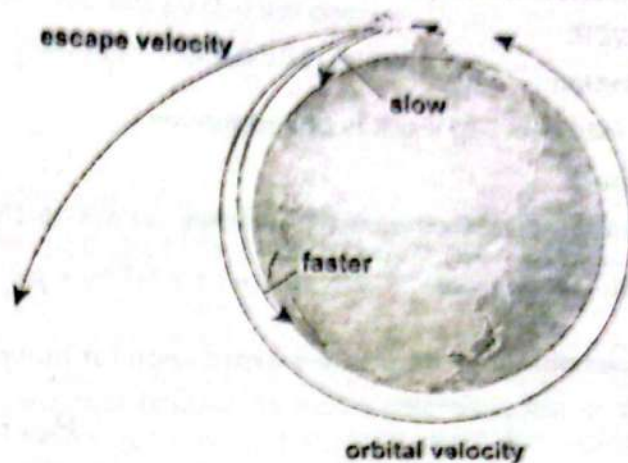
ESCAPE VELOCITY:

Escape velocity is the speed that an object needs to be traveling to break free of a planet or moon's gravitational field and leave it without further propulsion.

Derivation:

The simplest way of deriving the formula is by using the concept of conservation of energy. Let us assume that the object is trying to escape from a planet (which is uniform circular in nature) by moving away from it.

The prime force acting on such an object will be the planet's gravity. As we know, Kinetic energy (K) and the Gravitational Potential Energy (U_g) are the only two kinds of energies associated here.



By the principle of conservation of energy, we can write:

$$(K + U_g)_i = (K + U_g)_f$$

Where,

$$K = \frac{1}{2}mv^2$$

$$U = \frac{GMm}{r}$$

Here U_{gf} is zero as the distance is infinity and K_f will also be zero as final velocity will be zero.

Thus, we get:

$$\frac{1}{2}mv_{esp}^2 - \frac{GMm}{r} = 0 + 0$$

$$\frac{1}{2}mv_{esp}^2 = \frac{GMm}{r}$$

$$\Rightarrow v_{esp} = \sqrt{\frac{2GM}{r}}$$

The minimum velocity required to escape from the gravitational influence of massive body is given by:

$$v_{esp} = \sqrt{2gr}$$

Where,

$$g = \frac{GM}{r^2}$$

Chapter

5

Power

POWER:**Definition**

The rate of doing work is called power.

Formula

When an amount of work ΔW is done in time Δt , the average power P_{av} is defined as

$$P_{av} = \frac{\Delta W}{\Delta t}$$

We can obtain an alternative expression for power as

$$P_{av} = \frac{F \cdot S}{t}$$

If W is the work done when a constant force of magnitude F , points in the direction of the displacement " i.e. $\theta = 0$ Then

$$P_{av} = \frac{FS \cos \theta}{t}$$

$$P_{av} = \frac{FS \cos(0)}{t}$$

$$P_{av} = \frac{F \cdot S}{t}$$

$$P_{av} = F \cdot \frac{S}{t}$$

$$P_{av} = P = F \cdot V$$

It has been proved that power is the dot product of force and velocity

Units Of Power

- In S.I units, the unit, of power is watt (W), which is equal to J/Sec.

$$1W = 1J / 1s$$

- The multiples of watt are

$$1 \text{ mega watt} = 1 \text{ MW} = 10^6 \text{ W}$$

$$1 \text{ giga watt} = 1 \text{ GW} = 10^9 \text{ W}$$

- In British engineering system, the unit of power is ft.lb/sec. (Foot. Pound / Second)
- A bigger unit of power is called horsepower.

$$1 \text{ horsepower} = 1 \text{ hp} = 746 \text{ watt.}$$

Conversion Of Kilo Watt – Hour Into Joule

One kwh is the energy delivered by the current in one hour when it supplies energy at the rate 1000 joules per second, i.e.

$$1 \text{ KWh} = 1 \text{ kilo watt} \cdot \text{hour}$$

$$1 \text{ KWh} = 1000 \text{ watt} \cdot 3600 \text{ sec}$$

$$1 \text{ KWh} = 1000 \text{ Joule/second} \times 3600 \text{ second}$$

$$1 \text{ KWh} = 3600000 \text{ Joule} = 3.6 \times 10^6 \text{ Joule}$$

It is a scalar quantity having Dimensions $[ML^2T^{-3}]$

WORK DONE AGAINST FRICTION:

The work done against friction always results in a loss of kinetic energy. This is because the frictional force acts in the opposite direction of the motion of the object, and it does work to slow down the object. The lost kinetic energy is converted into heat, which is dissipated into the environment.

For example, when you walk, you are doing work against friction to move your feet forward. The frictional force between your shoes and the ground slows down your feet, and the lost kinetic energy is converted into heat. This is why your feet get warm when you walk for a long time.

Work Done Against Friction Is Dissipated as Heat In The Environment:

No system is perfect. Energy is dissipated whenever there is a change in a system.

OR

Wasted energy is the byproduct of any energy conversion process.

Energy is always lost when it is converted from one form to another. Here are some examples of electrical appliances they use electrical energy to work and waste energy.

Appliances	Useful energy	Wasted energy
Electric Kettle	The electrical energy that is used to heat the water.	The heat that is lost to the air around the kettle. The sound energy that is produced by the kettle when it is heating up
Hair Dryer	The heat that is used to dry your hair.	The heat that is lost to the air around the hair dryer. The sound energy that is produced by the hair dryer.
Light Bulb	The light that is emitted by the lightbulb	The heat that is emitted by the lightbulb.
TV	The images and sounds that are displayed by the TV.	The heat that is emitted by the TV.

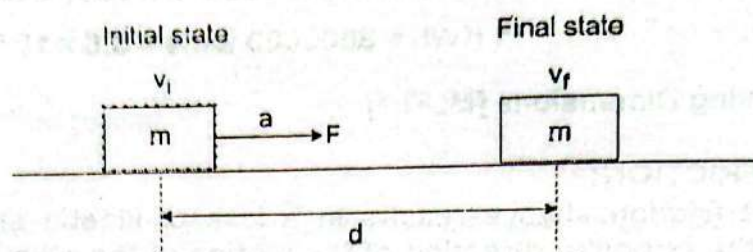
We can reduce wasted energy by making our appliances more efficient and by using them less. For example, we can use a more energy-efficient kettle, or we can only boil as much water as we need.

WORK ENERGY THEOREM:**Statement**

The work done by the net force on a particle equals the change in the particle's kinetic energy.

Mathematically:

$$W = \Delta K$$



Consider a body of mass m moving with initial velocity v_i after travelling through displacement d final velocity becomes v_f under the effect of force F .

As we know that

$$2ad = v_f^2 - v_i^2$$

$$a = \frac{v_f^2 - v_i^2}{2d}$$

hence external force acting on the body is

$$F = ma$$

$$F = m \frac{v_f^2 - v_i^2}{2d}$$

Therefore, work done on body by external force is

$$W = \vec{F} \cdot \vec{d}$$

$$W = m \frac{v_f^2 - v_i^2}{2d} \cdot d \cdot \cos(0)$$

$$W = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W = (K \cdot E)_f - (K \cdot E)_i$$

$$W = \Delta K$$

Chapter**5****Transformation of Energy****TRANSFORMATION OF ENERGY:****Statement**

Energy can neither be created nor it can be destroyed, but it can only be transformed from one form to another, the total energy remains constant.

Explanation:

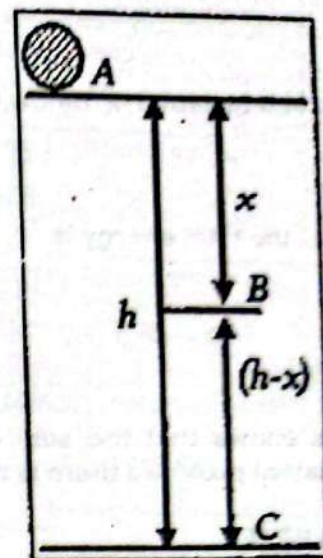
Energy cannot be created means one cannot produce energy by expanding nothing. Similarly we cannot destroy energy. We get some thing equivalent in return if we annihilate. It pair production is a good example of annihilation of energy. On the other hand in nuclear fission or fusion energy is created at the cost of mass. If 'm' is the mass annihilated, then according to Einstein's mass-energy relation the energy produced is

$$E = mc^2$$

Where 'c' is the velocity of light in vacuum

PROOF OF LAW OF CONSERVATION OF ENERGY:

With reference to the problem of a freely falling body, such as a body of mass 'm' placed at a point 'A' which is at a height 'h' from the surface of earth. The body possesses the P.E. equal to 'mgh' with respect to point 'C' lying at the surface of the earth. But the K.E of the body at point 'A' is zero. i.e.



AT POINT A

$$\text{Total Energy T.E} = \text{K.E} + \text{P.E}$$

$$\text{P.E} = 0 + mgh$$

$$\text{Total Energy at A} = mgh$$

AT POINT C

Now we calculate the kinetic energy at point 'C' for this we make the following data.

Initial velocity = $V_i = 0$ (at point 'P')

Final velocity = $V_f = V$ (at point 'O')

Acceleration = g

Distance = $S = h$

Now by using third equation of motion

$$2aS = V_f^2 - V_i^2$$

$$2gh = V^2 - (0)^2$$

$$V^2 = 2gh$$

Hence the kinetic energy of the body is

$$\text{K.E} = \frac{1}{2} mv^2$$

Put the value of v^2 in above equation

$$\text{K.E.} = \frac{1}{2} m. (2gh)$$

$$\text{K.E} = mgh$$

And at point 'C' the potential energy is taken arbitrarily equal to zero.

$$\text{Total Energy at point B.} = \text{K.E.} + \text{P.E}$$

$$\text{Total Energy at point B.} = mgh. + 0$$

We now calculate the potential energy and kinetic energy at any point 'B' at a distance 'x' below the point 'A'

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AT POINT B

$$P.E = mg(h - x)$$

$$K.E = \frac{1}{2}mv^2$$

&

But at a distance 'x' below the point 'p' the K.E will be

$$K.E = \frac{1}{2}mv^2$$

$$K.E = \frac{1}{2}m(2gx)$$

$$K.E = mgx$$

Thus the total energy is

$$\text{Total Energy at point B} = K.E + P.E$$

$$\text{Total Energy at point B} = mgx + mg(h - x)$$

$$\text{Total Energy at point B} = mgx + mgh - mgx$$

$$\text{Total Energy at point B} = mgh$$

This shows that the sum of kinetic energy and the potential energy i.e. total energy is always constant provided there is no force of friction involved during the motion of the body.

EXAMPLES

Some most common examples of law of conservation of energy are

When we switch on our electric bulbs, their filaments are heated up and begin to emit light. In switching on the bulb we supply electrical energy to it. It is converted into heat and light energies. Here one form of energy (electrical) is converted in other forms (light and heat) of energies and the electrical energy supplied is equal to the sum of the heat energy and light energy and the energy is neither created nor destroyed.

In rubbing our hands we do mechanical work which produces an equal amount of heat energy, i.e.

$$\text{Mechanical energy} = \text{Heat energy} + \text{Losses.}$$

Chapter**5****Multiple Choice Questions**

MCQ'S		ANS.
1. Work done by centripetal force is always: (a) Maximum (b) Minimum (c) Zero (d) None of these		c
2. A body of mass 5 kg is moving with a momentum of 10 kg m/s. A force of 0.2 N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is: 1. 2.8 J (b) 3.2 J (c) 3.8 J (d) 4.4 J		d
3. The kinetic energy of a light and a heavy object is same. Which object has maximum momentum? (a) Light object (b) Heavy object (c) Both have same momentum (d) N.O.T		b
4. Two bodies of mass 1 kg and 2 kg have equal momentum. Then the ratio of their kinetic energies is: (a) 2:1 (b) 3:1 (c) 1:3 (d) 1:1		a

5. A body fallen from height h . After it has fallen a height $h/2$, it will possess : (a) only potential energy (c) half potential half kinetic energy	(b) only kinetic energy (d) more kinetic and less potential	c
6. Which of the following quantity can be calculated by multiplying force and velocity ? (a) acceleration	(b) power (c) torque (d) work	b
7. The minimum velocity given to an object so that it emerges out from the gravitational field of earth is about (a) 11.2 km/s	(b) 15.3 km/s (c) 5 km/s (d) 9.8 km/s	a
8. When one joule of work is done on a body in one second, power of body is said to be : (a) One watt	(b) 0.5 watt (c) zero (d) 100 watt	a
9. The absolute potential energy of an object depends on: (a) The object's mass and height (b) The object's mass and speed (c) The object's shape and size (d) The object's color and temperature		a
10. The escape velocity of a planet depends on which of the following factors? (a) The mass of the planet only (b) The radius of the planet only (c) Both the mass and the radius of the planet (d) The density of the planet		c

***** Exams Practice MCQs *****

1. The work done by the conservative force along a closed path is (a) Positive (c) Zero	(b) Negative. (d) None of these.	c
2. The dot product of force and velocity is (a) Work. (c) Energy.	(b) Power (d) Momentum.	b
3. In case of negative work, the angle between work the angle between the force and displacement is (a) Zero. (c) 180°	(b) 90° (d) 45°	c
4. If mass and speed both are double, then the kinetic energy of a moving body (a) Increase 4 times (c) Increase 8 times	(b) Increase 6 times (d) Increase 10 times	c

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5. Power is equal to	(a) $\frac{F \times \vec{d}}{t}$ (c) $\frac{F \cdot \vec{d}}{t}$	(b) $\frac{F \times \vec{V}}{t}$ (d) $\frac{\vec{F} \cdot \vec{V}}{t}$	c
6. Kilo-watt hour is the unit of	(a) Energy (c) Time	(b) Power (d) velocity	a
7. One horse power is equal to	(a) 400 watt (c) 746 watt	(b) 550 watt (d) 1055 watt	c
8. If $F = 4\hat{i} - 2\hat{j}$ and $d = 3\hat{i} + 4\hat{j}$ the work done will be	(a) 4 Joules (c) 12 joules	(b) 8 Joules (d) 2 Joules	a
9. Watt hour is the unit of	(a) Energy (c) Time	(b) Power (d) velocity	a
10. When the direction of force is opposite to the direction of displacement, then the work done will be,	(a) Positive (c) Zero	(b) Negative (d) None of these	b*
11. Which of the following is not the unit of power	(a) Kilowatt hour (c) Foot-lbs/sec	(b) Horse power (d) Joules/sec	a
12. The absolute P.E of a body of mass "m" in the earth's gravitational field is given by	(a) $\frac{-GM_e m}{r}$ (c) $\frac{-GM_e m}{R_e^2}$	(b) $\frac{GM_e m}{r}$ (d) $\frac{-GM_e m}{R_e^2}$	a
13. The work done in moving a body between two points in a conservative field is independent of	(a) Direction. (c) Path followed by a body.	(b) Force applied. (d) Power.	c
14. The dimension of power are	(a) ML^2T^{-3} (c) $M^2L^2T^{-1}$	(b) $M^2L^2T^{-3}$ (d) ML^2T^2	a

15. The dimension of K.E are (a) $1/2 ML^2T^{-3}$ (c) $ML^{-1}T^{-2}$	(b) $1/2 M^2L^2T^{-3}$ (d) ML^2T^{-2}	d
16. Potential energy is increased when the work is done (a) Along the field. (c) By the field	(b) Against the field. (d) All of these.	b
17. The work done in lifting 30 kg of brick to a height of 20m is (a) 2800J (c) 5880J	(b) 600J (d) 2910J	c
18. Which of the following force is not the conservative (a) Coulombs force. (c) Frictional force.	(b) Force of a compressed elastic spring. (d) Gravitational force between two masses.	c
19. What is the direction of the force relative to the displacement? (a) Parallel (c) Perpendicular	(b) Antiparallel (d) Cannot be determined from the graph	a
20. When a body is connected to the earth, its potential is (a) Becomes positive. (c) Becomes zero.	(b) Becomes negative. (d) Remains unchanged.	c
21. For a variable force acting on an object, what does the area under the force vs displacement graph represent? (a) Distance travelled (b) Work done (c) Time (d) Change in velocity		b
22. Which of the following factors does the escape velocity of an object depend on? (a) The mass of the object (c) The distance of the object from the center of mass	(b) The radius of the object (d) All of the above	a
23. What is the escape velocity of an object from the surface of a black hole? (a) The speed of light (c) Less than the speed of light	(b) Greater than the speed of light (d) Cannot be determined	a
24. Which of the following planets has the highest escape velocity? (a) Mercury (b) Venus (c) Earth (d) Jupiter		d
25. What type of energy is the electric kettle using to heat the water? (a) Electrical energy (c) Light energy	(b) Heat energy (d) Sound energy	a

Chapter 5

Chapter Formulae & Numericals

Work done by constant Force	$W = \vec{F} \cdot \vec{S}$ $W = FS \cos \theta$
Work done by variable Force	$W = \Sigma(\vec{F} \cdot \Delta \vec{d})$
Kinetic Energy	$K.E = \frac{1}{2}mv^2$
Gravitational Potential Energy	$P.E = mgh$
Absolute Potential Energy at the surface of the Earth	$(P.E)_{abs} = \frac{-GmM_e}{R_e}$
Absolute Potential energy At certain Height	$(P.E)_{abs} = \frac{GmM_e}{R_e} \left(1 - \frac{h}{R_e}\right)$
Escape Velocity	$V_{es} = \sqrt{2gR}$
Power	$P = \frac{W}{t}$ $P = \vec{F} \cdot \vec{V} = Fv \cos \theta$
Work Energy Theorem	$W = \Delta K$